Exercises 11

Exercise 1 (Euclidean algorithm)**.** *Recall that the* greatest common divisor $\gcd(a, b)$ *of integers a and b is the largest positive integer that divides both a and b. Assuming (without loss of generality) that $a \geq b$, note that*

$$\gcd(a, b) = \gcd(a - b, b).$$

More generally, you can subtract any multiple of the smallest number from the biggest number without affecting the value of their gcd*:*

$$\gcd(a, b) = \gcd(a - mb, b),$$

for any $m \geq 0$. To decrease the bigger number by as much as possible, it is useful to choose m as big as possible while still making sure that $a - mb \geq 0$. This corresponds to $m = \lfloor a/b \rfloor$ (or `m = a//b` *in Python). With this choice $a - mb$ is the remainder of a modulo b (or* `a % b` *in Python). You can thus find the* gcd *by repeatedly replacing the largest number by itself modulo the smallest one. This is exactly how the* Euclidean algorithm *works.*

1. *Let $a = 210$ and $b = 91$. Compute $d = \gcd(a, b)$.*
2. *Find integers x and y that satisfy the* Bézout's identity

$$ax + by = d \tag{1}$$

using the extended Euclidean algorithm *(it is similar to the regular Euclidean algorithm, but you also have to keep track of the quotients* `a//b` *at all steps).*

Exercise 2 (Multiplicative inverse and order)**.** *Let $\alpha = a/d$ and $\beta = b/d$, where a, b, d are the values from the previous exercise. Note that $\gcd(\alpha, \beta) = 1$, i.e., α and β are* relatively prime*.*

1. *Find the* modular multiplicative inverse *of β modulo α, i.e., find an integer y such that*

$$\beta y \equiv 1 \pmod{\alpha}.$$

Hint: *Divide both sides of eq. (1) by d and then take them modulo α.*

2. *Determine the* multiplicative order *of β modulo α, i.e., find the smallest $r \geq 1$ such that*

$$\beta^r \equiv 1 \pmod{\alpha}.$$

You can do this by computing

$$\beta^k \pmod{\alpha},$$

for $k = 0, 1, 2, \ldots$, until you get 1.

3. *Determine the modular multiplicative inverse of* 19 *modulo* α.

Exercise 3 (Using order finding to factor)**.** *Your goal is to factor* $N = 21$ *using order finding. Assume the random number* $a \in \{2, \ldots, N-1\}$ *you picked in the first step of the algorithm is:*

1. $a = 7$,
2. $a = 10$.

Go through the remaining steps of the algorithm and determine the factors of N*. (You can use a calculator or a computer to compute* gcd *and to find orders modulo* N*.)*

Exercise 4 (`Cirq`: Quantum Fourier transform)**.**

Exercise 5 (`Cirq`: Phase estimation)**.**

Exercise 6 (`Cirq` (bonus): Order finding)**.**